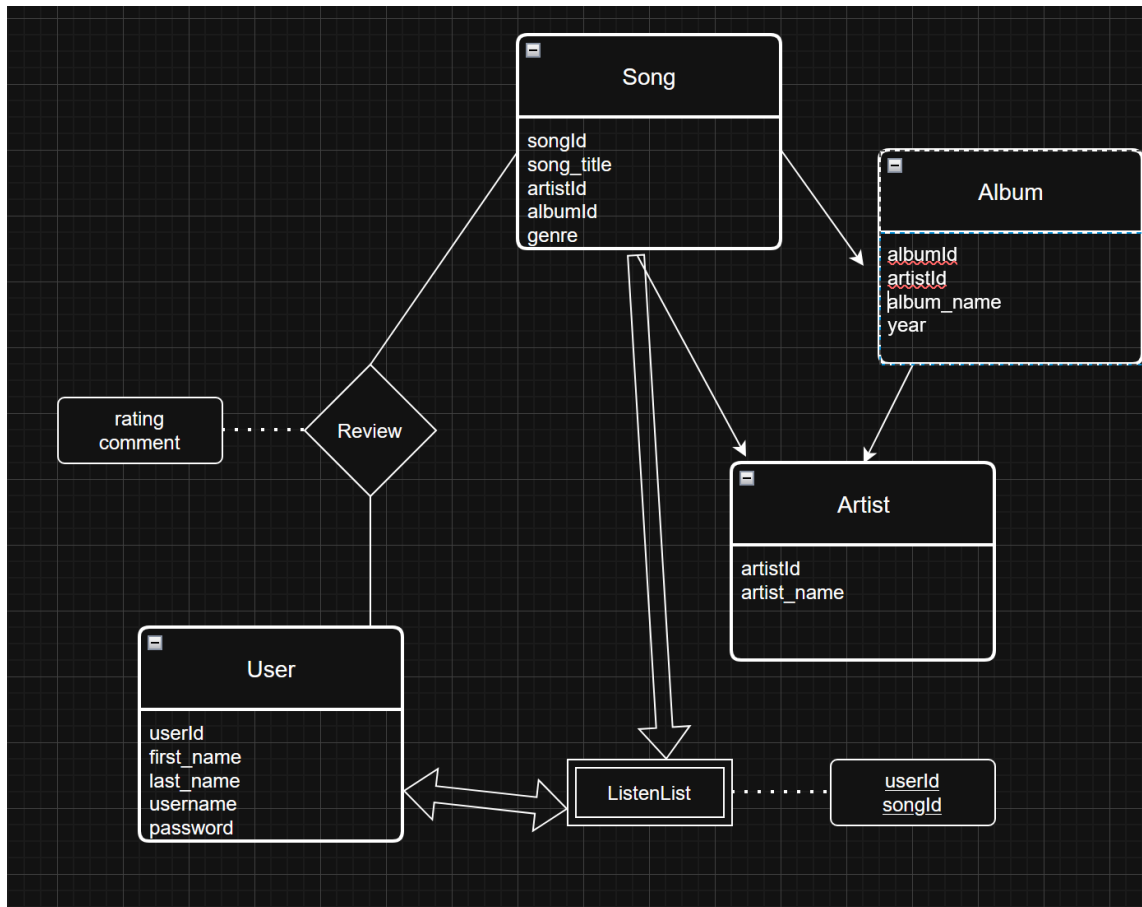


ER Diagram and tables:



Review
 userID (FK)
 songID (FK)
 rating
 comment
 (userID, songID) (PK)

Song
 songID (PK)
 song_title
 artistID (FK)
 albumID (FK)
 genre

Album
 albumID (PK)
 artistID (FK)
 year
 album_name

User
 userID (PK)
 first_name
 last_name
 username
 password

Artist
 artistID (PK)
 artist_name

ListenList
 userID (FK)
 songID (FK)
 (userID, songID) (PK)

Normalization using BCNF:

$R = \{ \text{songID}, \text{song-title}, \text{artistID}, \text{albumID}, \text{genre}, \text{userID}, \text{password}, \text{first-name}, \text{last-name}, \text{username}, \text{password}, \text{album-name}, \text{year}, \text{artist-name}, \text{rating}, \text{comment} \}$

FD:
 $\text{songID} \rightarrow \text{song-title}, \text{artistID}, \text{albumID}, \text{genre}$
 $\text{userID} \rightarrow \text{first-name}, \text{last-name}, \text{username}, \text{password}$
 $\text{albumID} \rightarrow \text{artistID}, \text{year}, \text{album-name}$
 $\text{artistID} \rightarrow \text{artist-name}$
 $\text{songID}, \text{userID} \rightarrow \text{rating}, \text{comment}$

$\text{songID}^+ = \{ \text{songID}, \text{song-title}, \text{genre}, \text{artistID}, \text{albumID}, \text{album-name}, \text{year}, \text{artist-name} \}$

$R - \text{songID}^+ + \text{songID} = \{ \text{userID}, \text{password}, \text{first-name}, \text{last-name}, \text{username}, \text{password}, \text{rating}, \text{comment}, \{ \text{songID} \} \} = R_2$

$\text{userID}^+ = \{ \text{userID}, \text{first-name}, \text{last-name}, \text{username}, \text{password} \}$

$R_2 - \text{userID}^+ + \text{userID} = \{ \text{userID}, \text{rating}, \text{comment}, \text{songID} \}$

Tables:

songID (pk) song-title genre artistID albumID album-name Year artist-name	userID (pk) first-name last-name username password	userID songID rating comment ($\text{userID}, \text{songID}$) (pk)
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The differences in the tables arises due to BCNF using closure of attributes as determined by the FDs to create tables whereas the ER diagram looks at relationships between entities to create tables. We're using the tables created by the ER diagram because it creates less redundancy in terms of artist and album names and IDs. In the normalized tables, those attributes will be copied across all songs that share an artist or album. Using the ER tables eliminates this redundancy at a potential cost of higher join

times. However, this cost is offset because the normalized song table has larger rows and separate indices may need to be created to access albumId and artistId efficiently and those indices also take up more room.